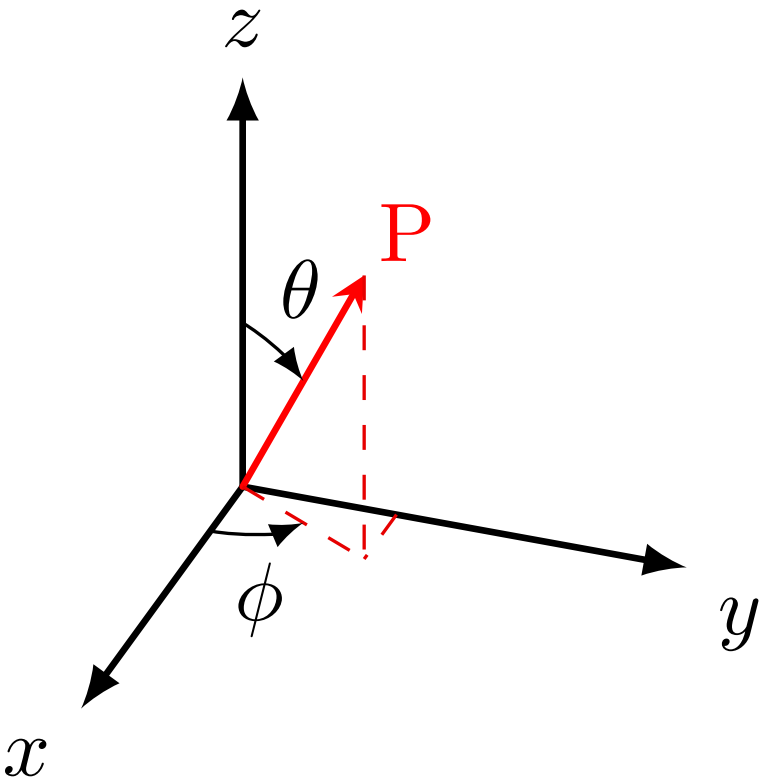
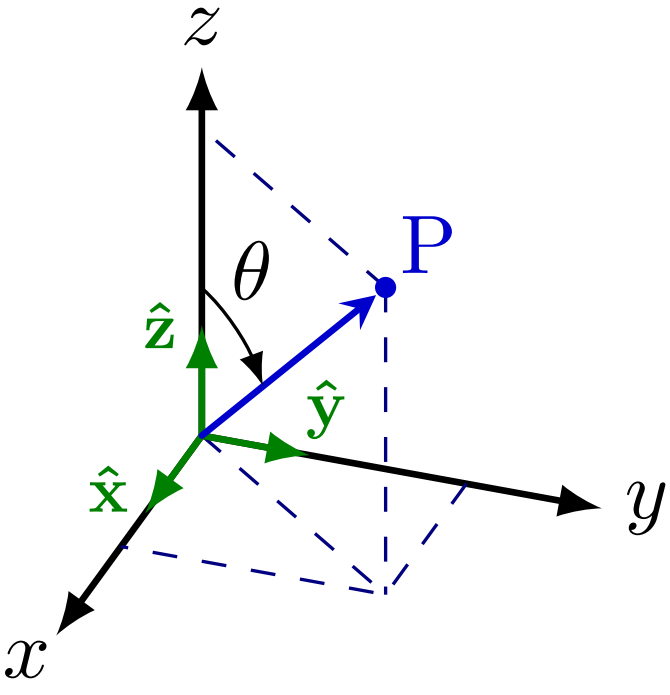
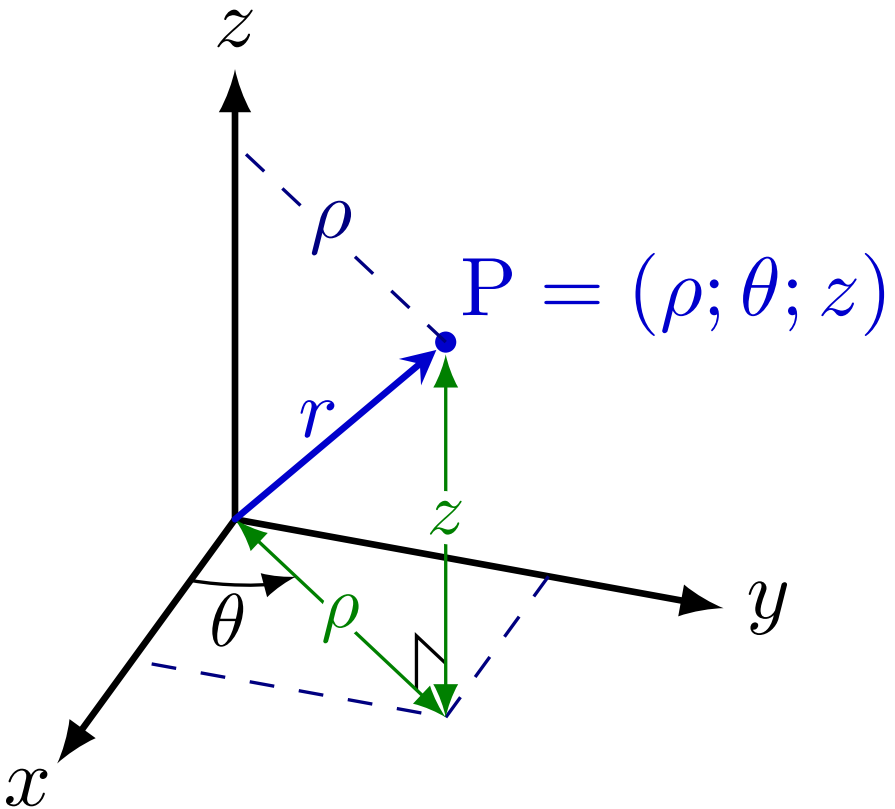
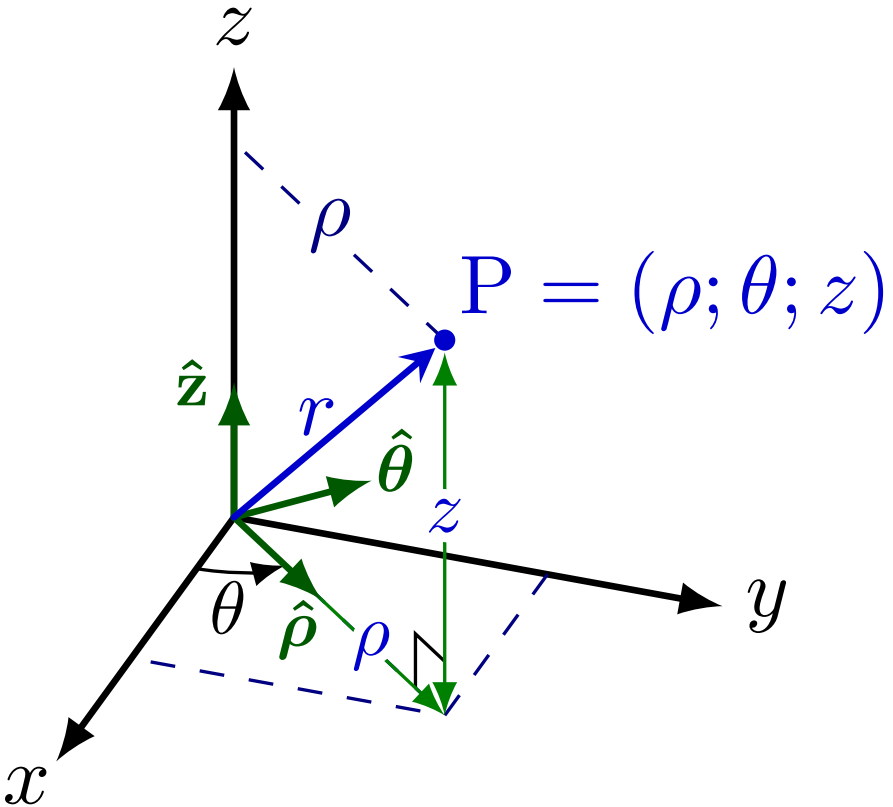
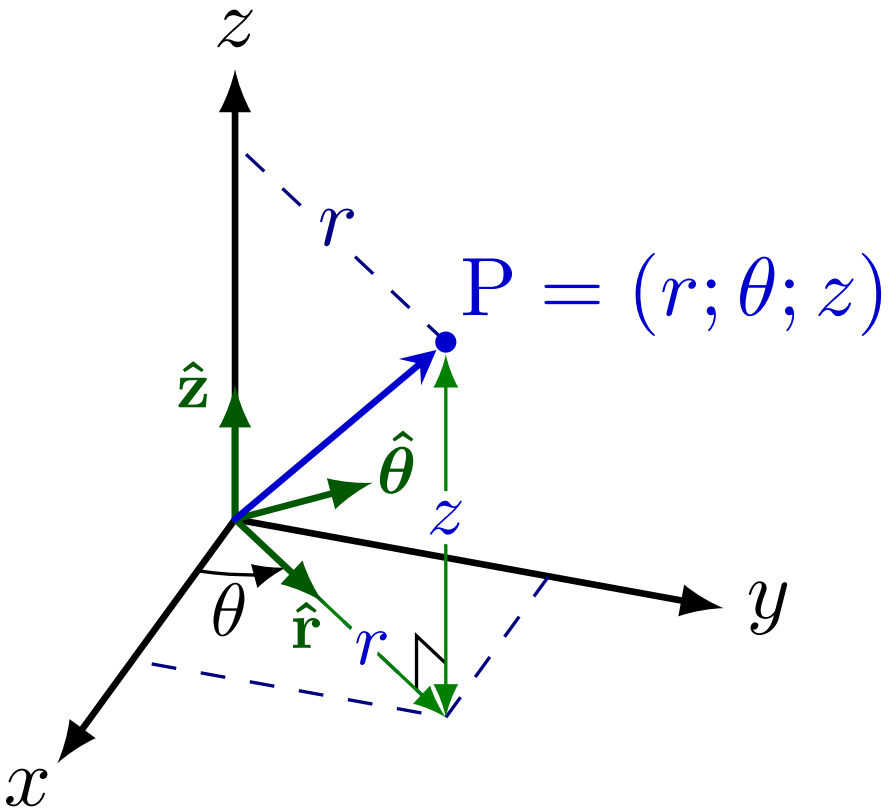
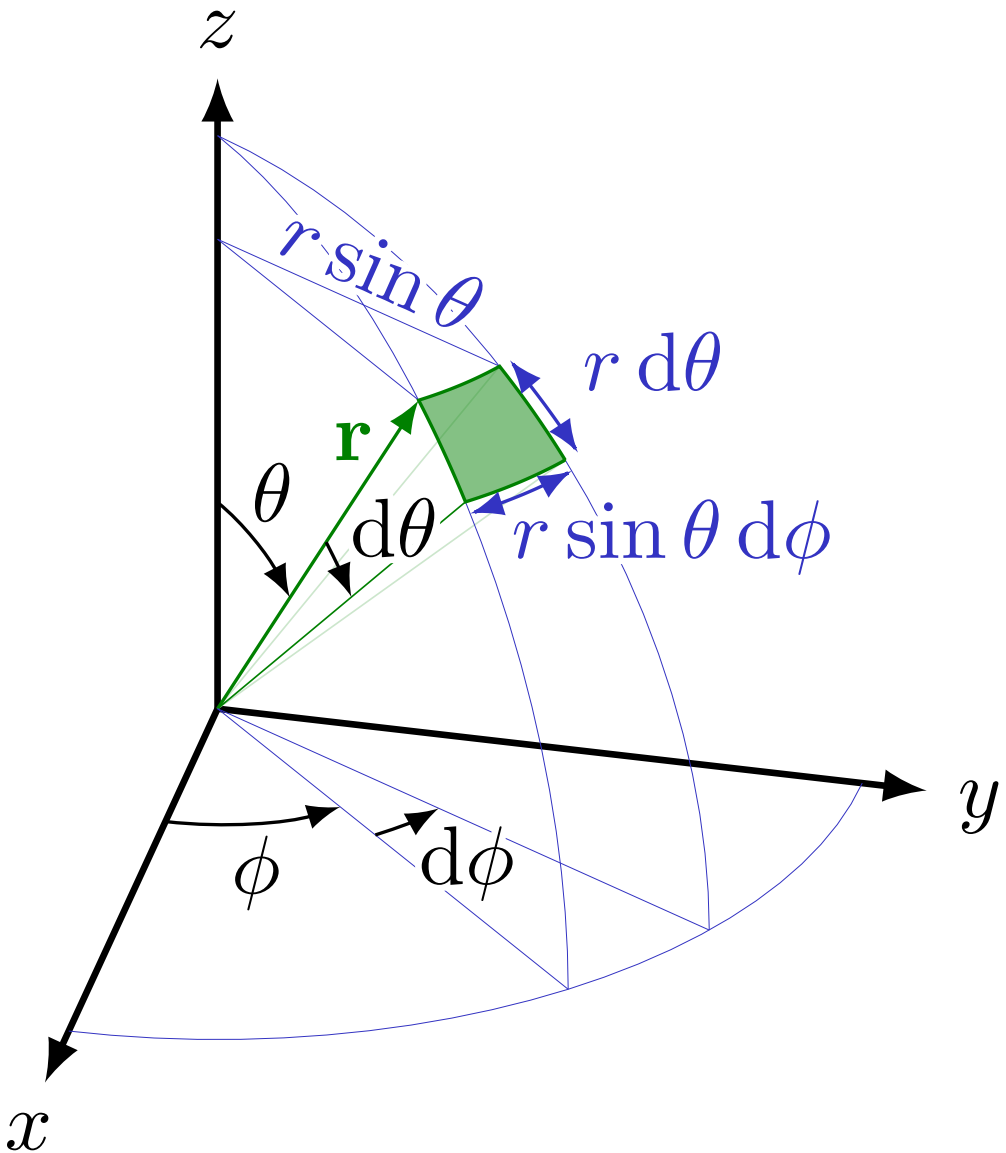

z
P
θ
ϕ
y
x

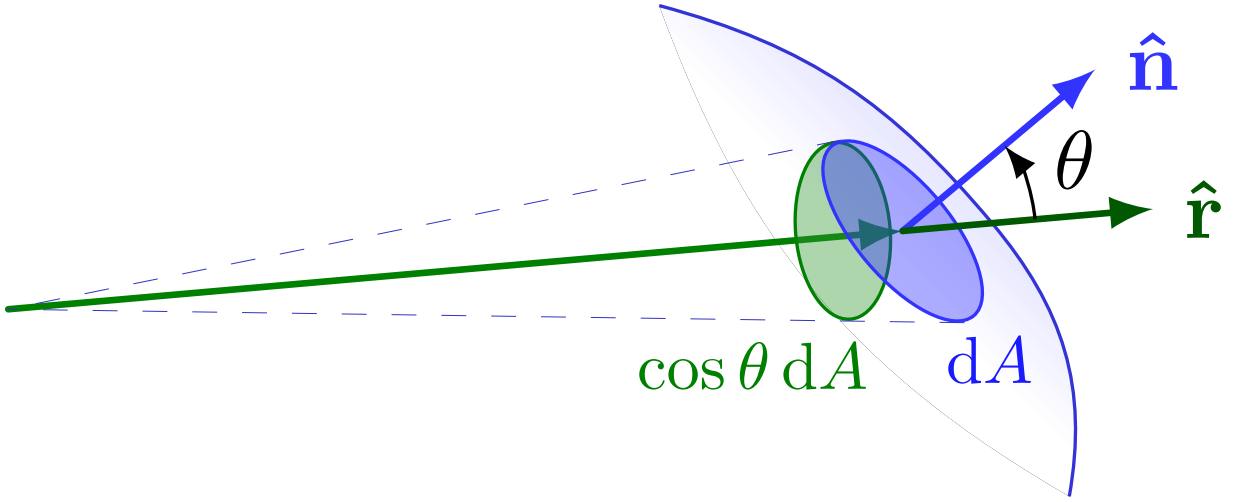

z
P
θ
ẑ
ŷ
x̂
y
x


z
ρ
P = (ρ; θ; z)
r
z
θ
ρ
y
x


z
ρ
P = (ρ; θ; z)
ẑ
r
θ̂
z
θ
ρ̂
ρ
y
x


z
r
P = (r; θ; z)
ẑ
θ̂
z
θ
r̂
r
y
x


z
$r\sin\theta$
$r\,\mathrm{d}\theta$
$\mathbf{r}$
θ
$\mathrm{d}\theta$
$r\sin\theta\,\mathrm{d}\phi$
y
ϕ
$\mathrm{d}\phi$
x


n̂
θ
r̂
cos θ dA
dA